Network Access Control

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Agenda

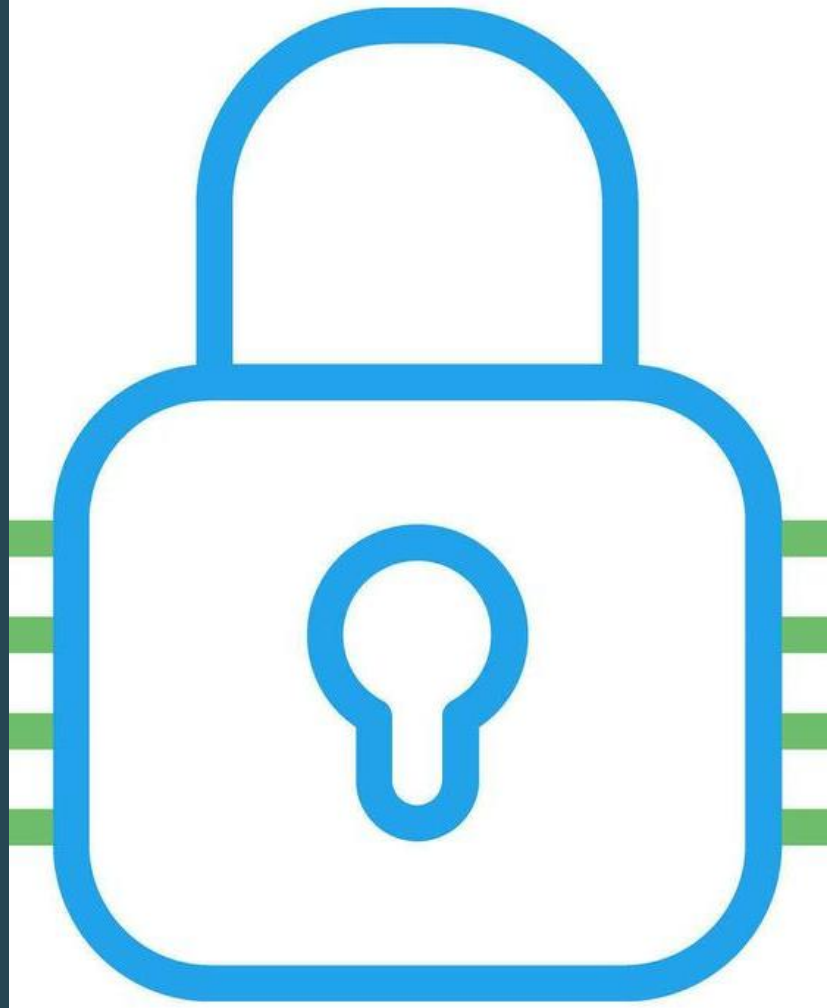
1. Definition
2. Elements of a NAC System
3. Enforcement Methods
4. Extensible Authentication Protocol
5. IEEE 802.1X

Definition

NAC

An umbrella term for managing access to a network

- Authenticates users logging into the network
- Determines what data they can access
- Determines what actions they can perform
- Examines health of user's computer/mobile device



Elements of a NAC System

Access Requestor

- Node attempting to access the network
- Workstations, servers, printers, cameras, other IP-enabled devices.
- Supplicants/Clients

Policy Server

- Determines what access should be granted
- Access based on host/AR's posture/condition
- Condition determined using backend systems – antivirus, patch management, user directory,

Network Access Server

- Access control point for users in remote locations connecting to enterprise's network
- Media gateway/Remote Access Server (RAS)
- Can also be a Policy Server

Generic Network Access Diagram

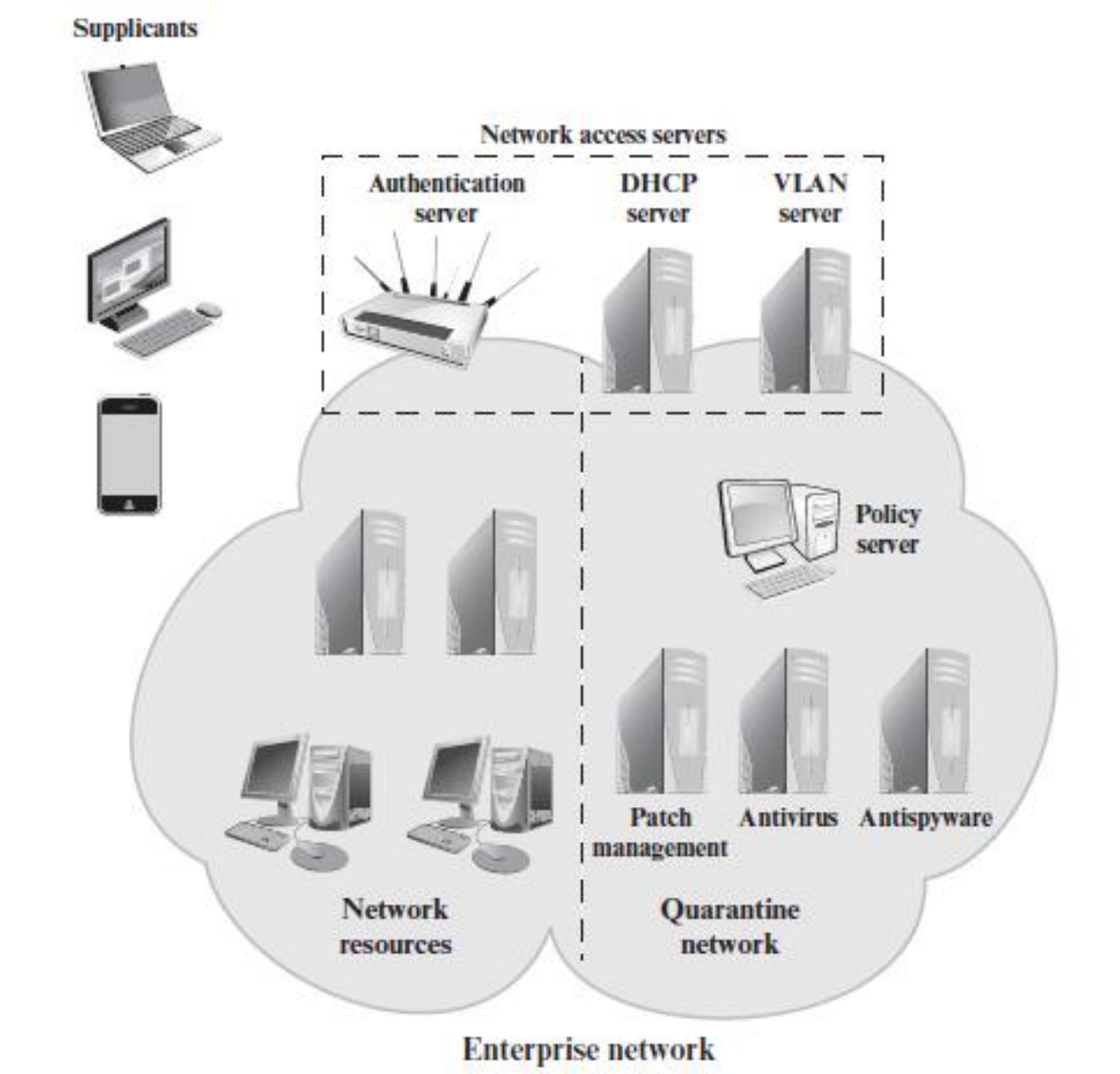
ARs seek access to enterprise network by applying to NAS.

Authenticate the AR → secure protocol + cryptographic keys

Verify identity, give access privileges, establish session keys

Health, suitability, screening, and assessment checks on AR

Acceptable Authorization Credentials + Health Check Passed → Remote Access; else denied/limited access

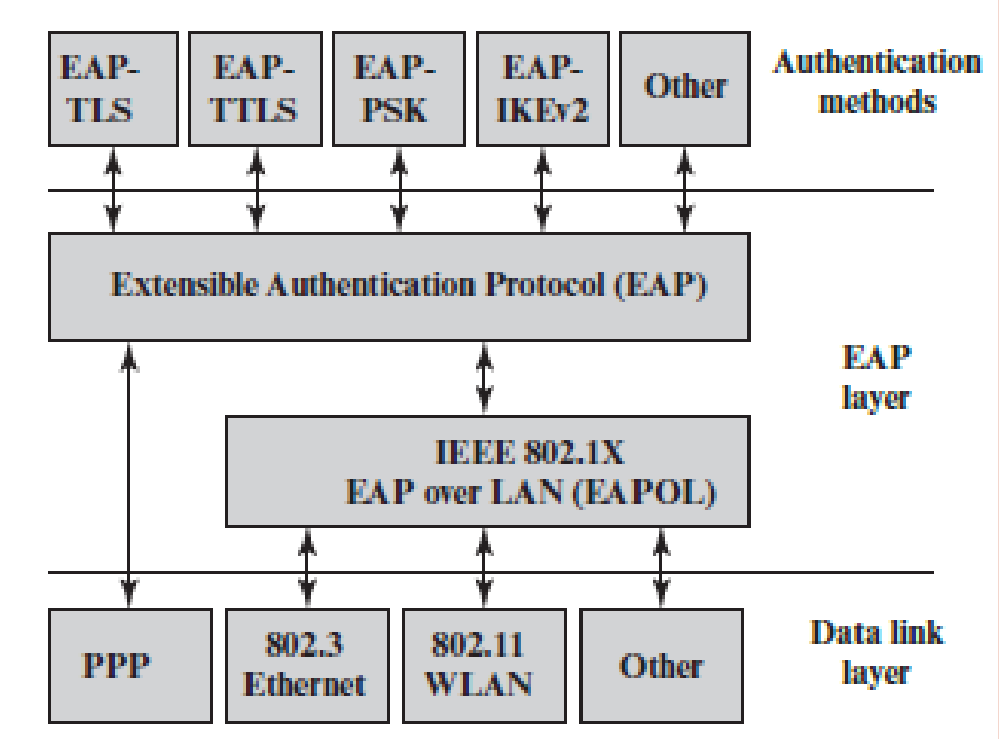


Network Access Enforcement Methods

IEEE 802.1X	• Link Layer protocol • Enforces authorization before a port is assigned an IP address • Makes use of Extensible Authentication Protocol
VLANs	• Enterprise network consists of an interconnected set of LANs • VLAN – logical subgroup within a LAN; created dynamically via S/W rather than moving cables • NAC decides which of the network's VLANs it will direct an AR towards
Firewall	• Provides a form of NAC • Allows/denies network traffic between an enterprise host and external user
DHCP Management	• Dynamic Host Configuration Protocol – enables dynamic allocation of IP addresses to hosts • DHCP server intercepts DHCP requests, and assigns IP addresses • So, NAC enforcement occurs at IP layer, based on subnet and IP assignment

Extensible Authentication Protocol

- Defined in RFC 3748
- Framework for network access and authentication protocols
- Provides a set of protocol messages which encapsulates authentication methods
- Can operate a variety of network and link-level facilities – point-to-point links, LANs



Authentication Methods

EAP provides a generic transport service for the exchange of authentication information.
Multiple authentication methods supported, any specific one used by EAP client and authentication server.

EAP-TLS

Transport Layer Security

- Client, server authenticate each other using digital certificates – dual-layer certification.
- Unmatched security – safeguards against man-in-the-middle attacks.
- High operational complexity – managing a comprehensive public key infrastructure.
- Distributing, renewing, and revoking digital certificates requires significant resources

EAP-TTLS

Tunneled TLS

- More adaptable solution than EAP-TLS.
- Secure, encrypted tunnel/communication channel between client and server created.
- Password Authentication Protocol (PAP)/ Challenge-Handshake Authentication Protocol (CHAP) then used to transmit and validate user's credentials.

Authentication Methods

EAP-GPSK

Generalized Pre-Shared Key

- Mutual authentication and session key derivation using PSK.
- Provides protected communication channel after mutual authentication.
- Efficient in terms of message flows, computational costs.
- Requires existence of pre-shared keys between client and server.

EAP-IKEv2

Internet Key Exchange v2

- Supports mutual authentication and session key establishment.
- IKEv2 – exchange of messages in pairs.

Successful Authentication – an exchange of EAP messages, as a result of which the authenticator decides to allow access to the peer, and the peer decides to use this access.

Goal of exchange of EAP Messages

EAP Exchanges

Remote Authentication Dial-In User Service

Peer

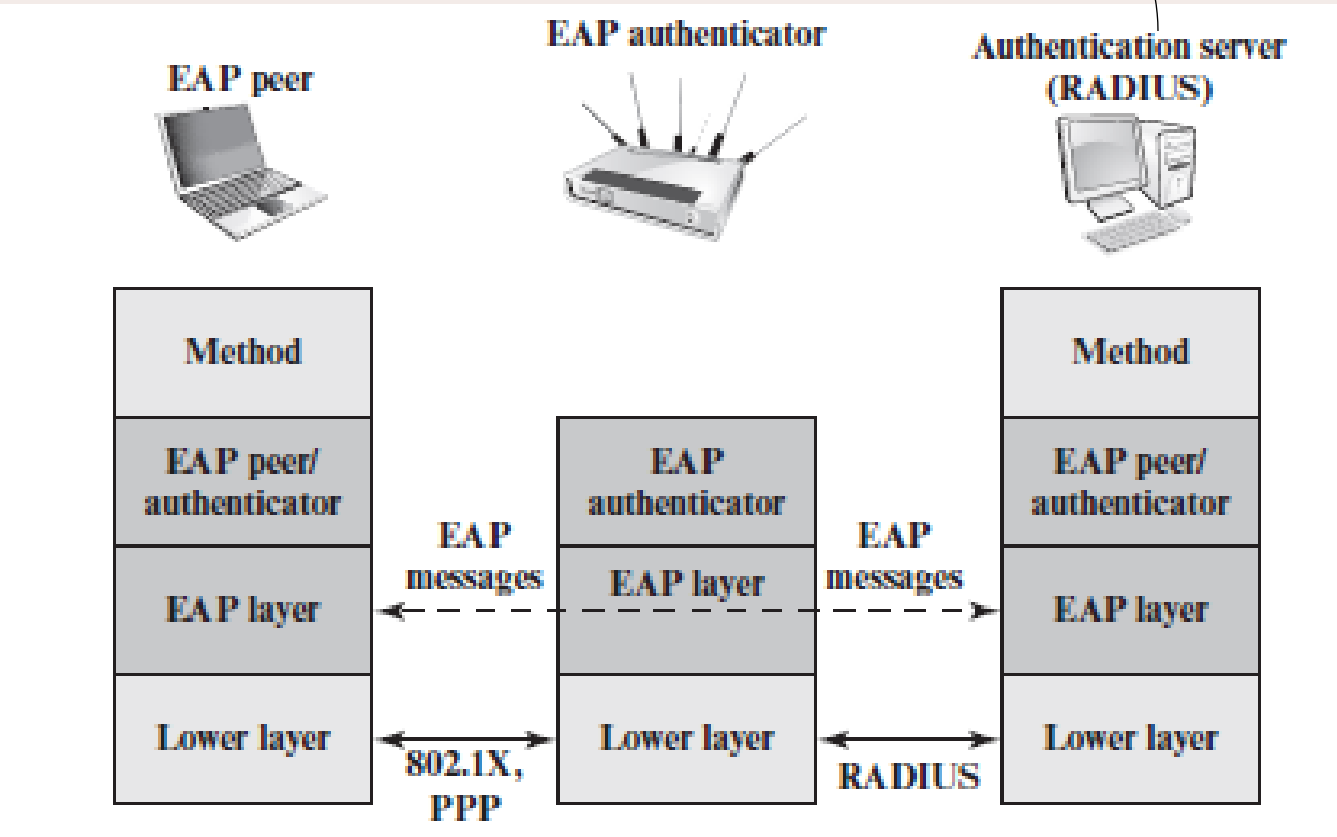
- Client computer trying to access network.

Authenticator

- Access point/NAS – requires EAP authentication before granting access

Authentication Server

- Negotiates use of EAP method with peer
- Validates peer, authorizes access



EAP Message Fields

Code

- Identifies the type of EAP message.
- Request (1), Response (2), Success (3), Failure (4)

Identifier

- Used to match responses with requests.

Length

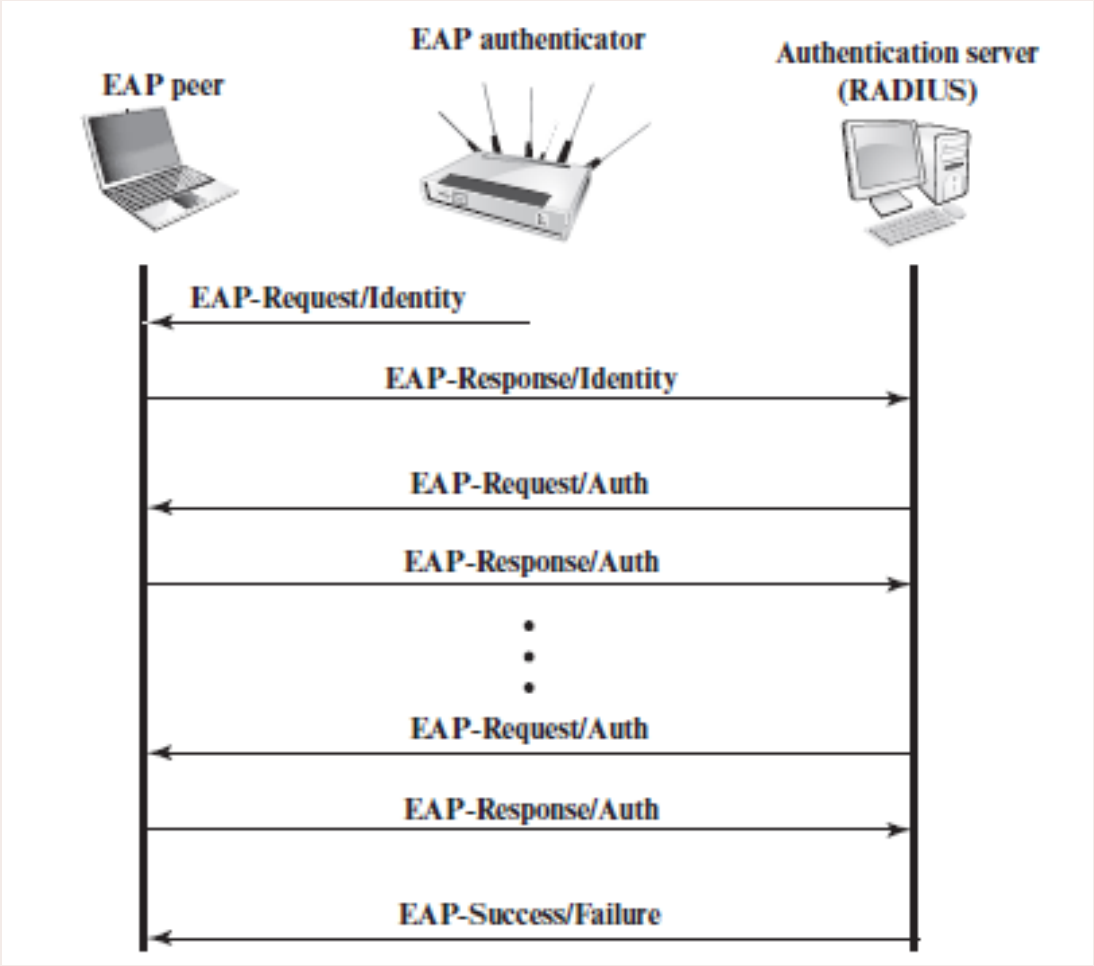
- Indicates length of EAP message in octets.
- Includes code, identifier, length, and data fields.

Data

- Contains information related to authentication, plus a Type field.
- Success and Failure messages don't have a Data field.

Pass-Through Mode

- Peer sends message to authenticator to request EAP exchange.
- First pair of Request and Response messages is of Type *Identity*.
- Next, server selects EAP method, sends message with Type *Auth* method.
- If peer supports EAP method, it sends Response of same type, else NAK.
- EAP method determines no. of Request/Response pairs.
- Authentication info, including key material, is exchanged.
- Either authentication **success** or **failure**.



IEEE 802.1X Port-Based NAC

Authenticator
Entity at one end of a point-to-point LAN segment, facilitates authentication of entity to the other end of the link.

Authentication Exchange
Two-party convo between systems performing authentication process.

Authentication Process
Cryptographic operations and supporting data frames that perform authentication.

Authentication Server
Provides authentication service; Determines whether supplicant is authorized

Authentication Transport
Datagram session – actively transfers authentication exchange between two systems

Bridge Port
Port of IEEE 802.1D or 802.1Q bridge.

Edge Port
Bridge port attached to a LAN that has no other bridges attached to it.

Network Access Port
Point of attachment of a system to a LAN. Can be physical or logical (IEEE 802.11).

Port Access Entity
Protocol entity associated a port.

Supplicant
Entity that seeks to be authenticated.

EAP over LAN – EAPoL

What?

Essential element in 802.1X.

Operates at network layers.

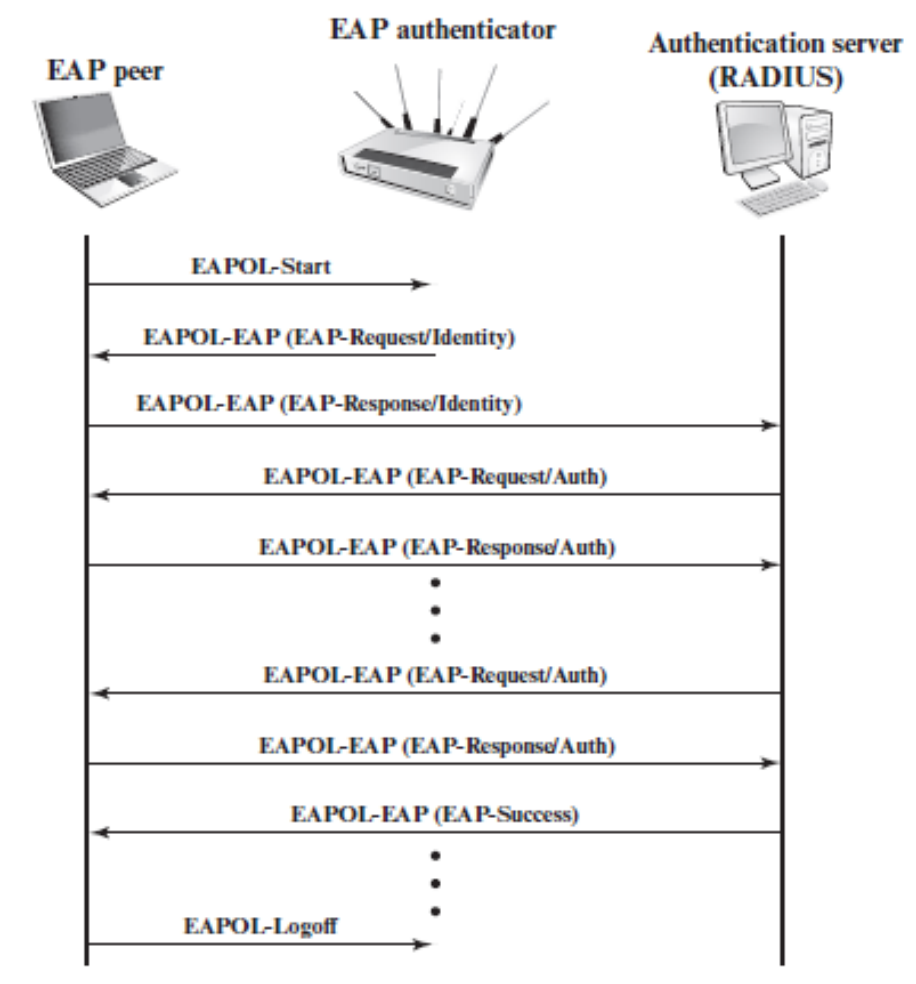
Enables supplicant to communicate with authenticator, supports exchange of EAP packets for authentication.

Frame Types

- EAPoL-EAP
- EAPoL-Start
- EAPoL-Logoff
- EAPoL-Key

Packet Format

- Protocol Version
- Packet Type
- Packet Body Length
- Packet Body



EAPoL Process

Supplicant connects to LAN, doesn't know MAC address of authenticator, or if authenticator is even present.

Sends **EAPoL-Start** packet to determine if authenticator is present.

Authenticator sends identity message in **EAPoL-EAP** packet.

Authenticator sends cryptographic keys using **EAPoL-Key** once it has decided to admit supplicant to the network.

EAP-Logoff packet indicates that supplicant wishes to disconnect.

Thank You

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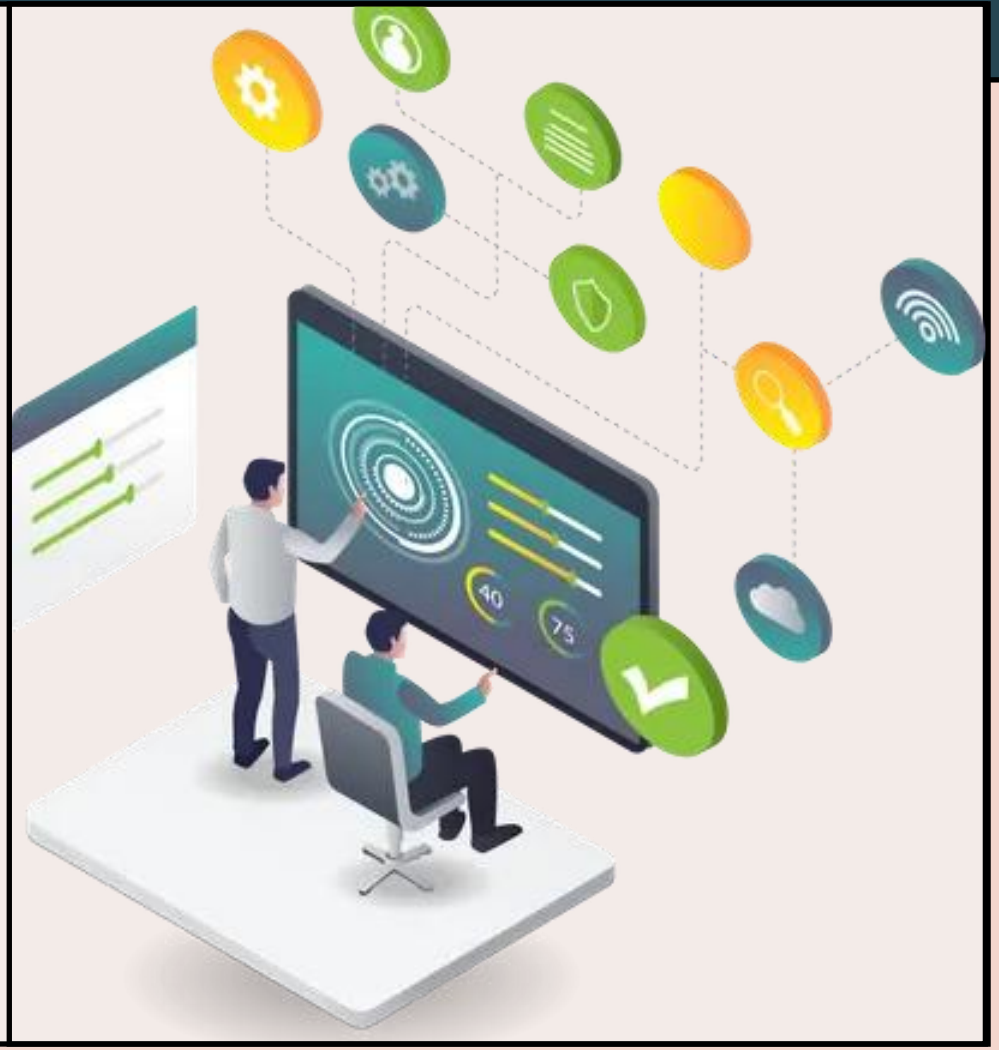
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